

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)	(i)		award (2) for all points plotted correctly (tolerance $\pm\frac{1}{2}$ square) award (1) for 3 / 4 points plotted correctly award (1) for suitable curve		3		3	2	1
		(ii)	I	15 cm ³ accept any value in the range 14-16 cm ³ allow ecf from graph			1	1		
			II	0.25 cm ³ /s (2) accept any value in the range 0.23-27 if incorrect award (1) for either of following dividing by 60 $\frac{15}{1}$ allow ecf from part I		2		2	2	
	(b)			award (1) each for any two of following - increase the temperature (of the acid) - increase the concentration (of the acid) - use smaller pieces (of magnesium) / increase the surface area (of the magnesium) / use (magnesium) powder neutral answer – catalyst	2			2		2